HYPERWAVE Systems Engineering

HSP ENGINEERING

PROJECT TITLE	:	**PUCHONG LOTING DOCUMENT STORAGE PROJECT**
OWNER	:	**HYPERWAVE SYSTEMS ENGINEERING**

DOCUMENT TITLE	:	**PUCHONG LOTING DOCUMENT STORAGE STRUCTURAL ASSESSMENT CALCULATION REPORT**
CLIENT DOCUMENT NUMBER	:	**HSE-HSP-ST-001**
STAGE	:	**DETAILED ENGINEERING**

PROFESSIONAL ENGINEER WITH PRACTISING CERTIFICATE
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MALAYSIA

00	16.04.21	ISSUED FOR APPROVAL	HSP	HSP		RT	
REV	**DATE**	**DESCRIPTION**	**PREPARED**	**ENDORSED**		**REVIEWED**	**REVIEWED**
			HSP			**HSE**	

OBJECTIVE

To perform structural assessment of the proposed document storage at Puchong Loting Hyperwave System Engineering office.

METHODOLOGY

1. Gather relevant existing design basis of the shop lot.
2. Estimate the document weight the storage may experience.
3. Model in STAADPro in 3D the proposed document storage platforms.
4. Apply all relevant primary load cases and load combinations per UBBL.
5. Assessed structure adequacy base on relevant provisions of BS BS5950-1:2000

ASSUMPTIONS AND LIMITATIONS

1. UBBL and SBO imposed load links to BS 6399 Table 1. For Offices for general use, the imposed uniformly distributed load is 2.5kN/m2.For stairs it is taken as 3kPa
2. Load combinations shall use BS code appropriate provisions.
3. The document storage housed inside the existing building. Thus, no windload effect will be considered.
4. Document weight shall be taken 25kg/m per layer (0.25kN/m/layer).
5. Wooden planks unit weight taken as 7kN/m^3.

REFERENCES

1. Uniform Building by-Laws 1984
2. BS 6399 Loading for Buildings
3. Steelwork Design Guide to BS 5950-1:2000

CONCLUSIONS

The proposedplatformframing,its connection and existing affected members is adequate to cater the intended design loadings.

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1.0 GENERAL

Hyperwave System Engineering proposed to install mezzanine floor as project document storage at their office building in Puchong Loting, Kuala Lumpur. The proposed document storage was made of 5 layers of pre-manufactured shelves.

1.1 SCOPE

This document covers the assessment of the proposed mezzanine platform to serve as project document storage.

1.2 DESIGN BASIS

The appropriate load combinations and structural calculation shall be referred BS 5950-1:2000 and engineering best practices.

1.3 MATERIALS

1.3.1 STEEL STRUCTURE SECTION AND PLATES

Structural steel shall be designed in accordance with BS 5950 (2000). Structural steel shall be grade S275 to BS EN 10025 and BS 5950. The following steel properties shall be used in the analysis:

Density	- 7850 kg/m^3
Modulus of Elasticity	- 210,000 MPa
Shear Modulus	- 8000 MPa
Poisson's Ratio	- 0.3

1.3.2 STRUCTURAL BOLTS

Structural bolts shall be BS 3692 Grade 8.8 in accordance with BS EN ISO 898-1, BS EN ISO 898-2.

1.3.3 WELDING ELECTRODE

Welding electrode yield strength equal (pw) to 360MPa, plate yield strength due to welding is equal to 220MPa per BS 5950-1:2000 Table 37.

1.3.4 ANCHOR BOLT

Post installed anchor shall be Hilti-mechanical anchor Class 5.8 or approved equivalent.

1.3.5 GROUT

Cement based non-shrink grout shall be used with minimum compressive strength of 75N/mm^2.

1.4 PLATFORM FRAMING GEOMETRY

The proposed mezzanine framing will be mounted into the stairwell of the existing building. All members are made of C100x50x10kg/m.

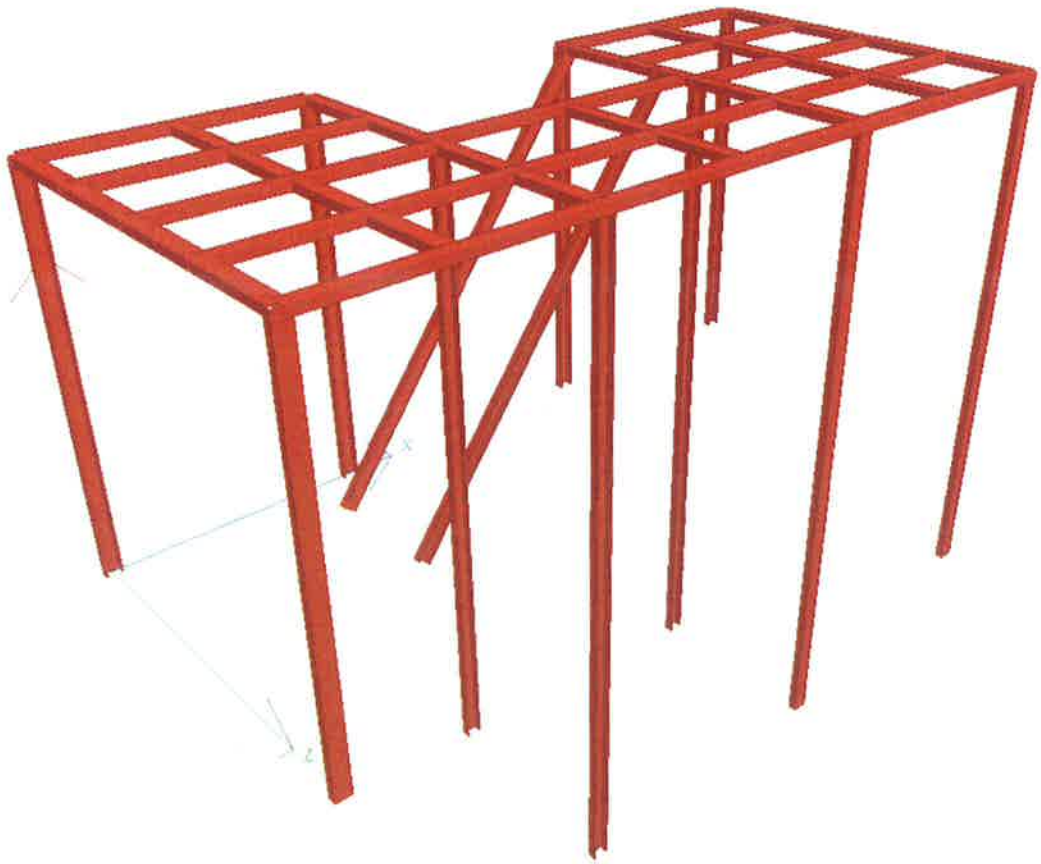


Figure 1. 3D Model of Mezzanine Framing Assembly

2.0 LOAD COMBINATIONS AND ESTIMATION OF LOADS

Load combinations shall follow the relevant clause of BS 5950-1:2000 Section 2.4, which was summarized with load factors as shown below Table 1. Only relevant load combinations are presented below:

Table1. ULS Load Combination

	without wind
Weight of structure	1.4
Live load	1.6

2.1 SLS LOAD COMBINATIONS

SLS load factor shall be taken as 1 for all loads.

2.2 DEAD LOAD

Dead load comprises of steel structure, handrail (55kg/m), 6mm thick grating (60kg/m2) and wooden plank (7kN/m^{3}x0.025m = 0.175kN/m^2).

2.3 LIVE LOAD

Live load shall be taken as 2.5kPa uniform load at no shelves & stair. While for loaded with shelves it can be estimated as 0.25kPa x 5 layers = 1.25kPa, which is lesser than the live load for stairs and unloaded area. Therefore, it is conservative to used 2.5kPa to the whole mezzanine floor. In reference to UBBL and SBO and eventually leads to BS6399, stair live load is taken as 3kPa. Thus, the mezzanine live load is lesser than the code recommended live load. Therefore, it can be considered that the mezzanine platform and its intended used can be taken by the existing building.

2.4 WIND LOAD

The proposed mezzanine floor is not exposed to weather. Therefore, wind load is not applied.

2.5 EARTHQUAKE LOAD

The mezzanine framing system is not part of the main seismic resisting frame of the whole building. As a component the proposed frames are fully supported by the existing building in which its perimeter post and beams are all anchored to the wall through M12 post-drilled anchor bolt. These anchors will effectively transfer shear forces into the main structure. The beam-columns & beam-beam connections are fully welded, which gives advantageous in reducing the slenderness factor of the elements.

Having the conditions above, it can be taken that the mezzanine framing will not take seismic forces. It will transfer seismic weight into the main frame instead. Therefore, the earthquake loading can be neglected.

3.0 STRUCTURAL ANALYSIS AND DEFLECTION ASSESSMENT

The structural analysis result utilization ratio has been presented herewith for simplicity. Detailed analysis shall be referred to Appendix A. The perimeter column effective length was taken to be 1m in calculating slenderness due to the presence of anchor studs into the wall.

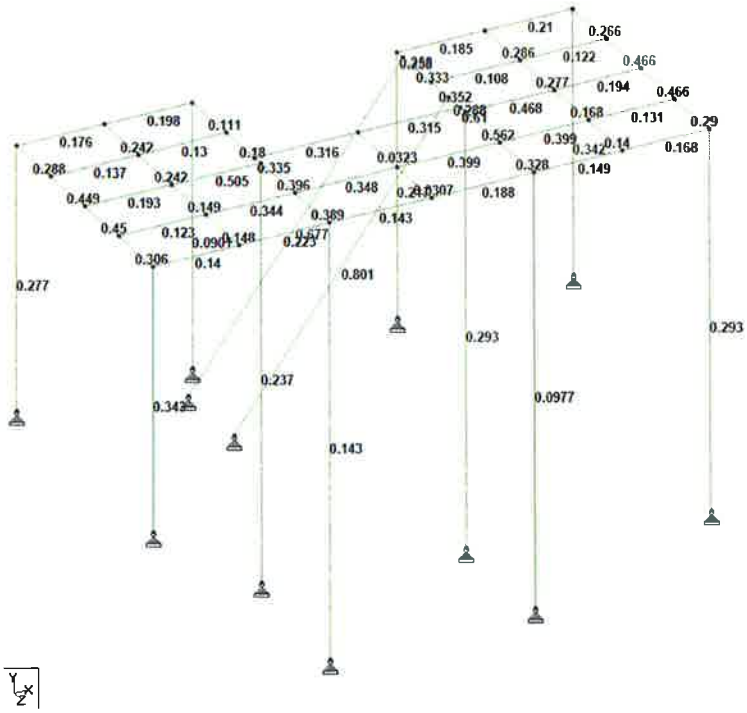


Figure 2. Utilization ratio are all less than unity. Therefore, the proposed framing is adequate.

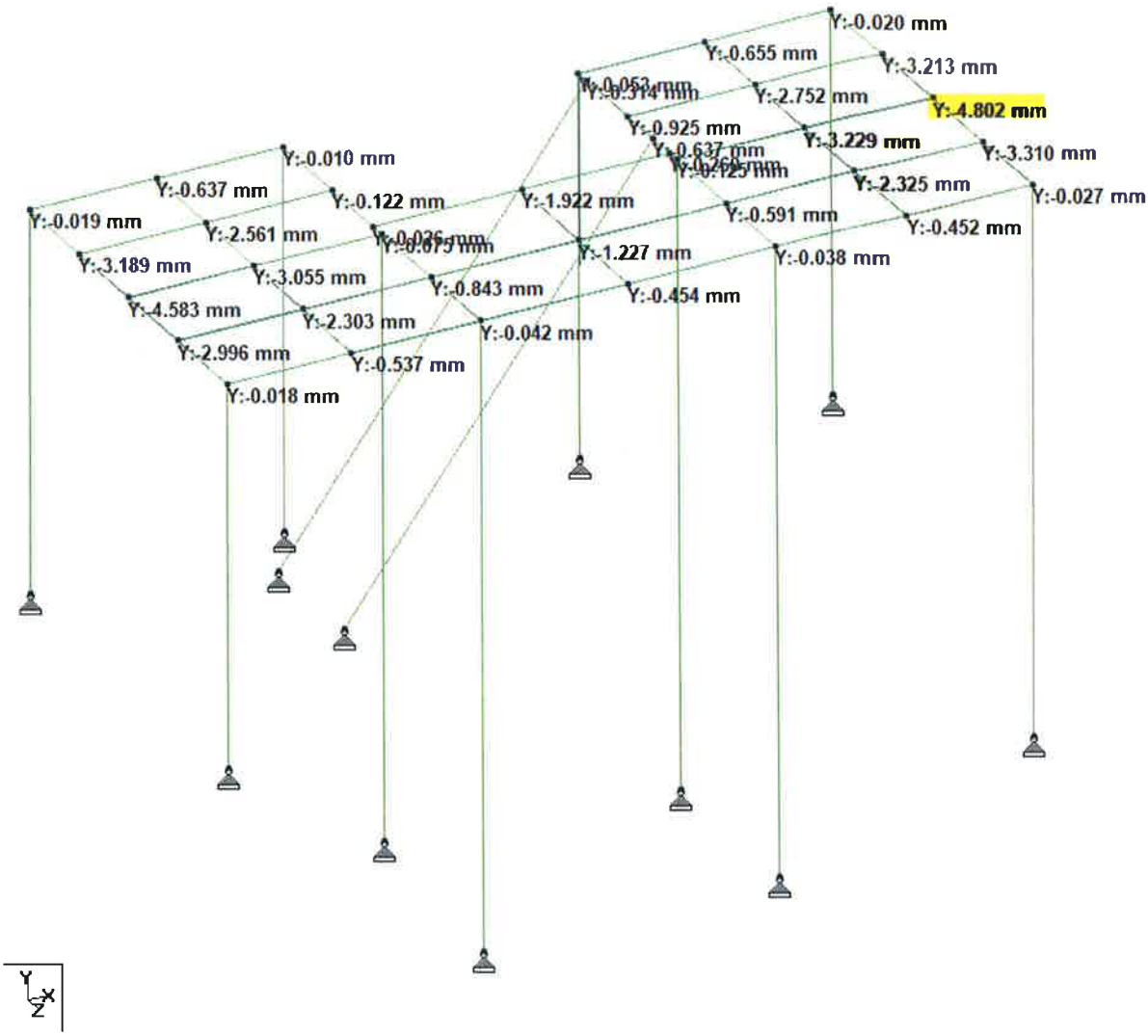


Figure 3. The deflection due to live load.

The maximum deflection due to live load was determined to be 4.80mm. In reference to Table 8 of BS5950 the allowable deflection due to live load can be taken as L/200 = 2400/200 = 12mm. Therefore, all members are adequate to cater the intended loads.

4.0 POST DRILL ANCHOR AND END PLATE ASSESSMENT

Each post base shall be provided with 2-M12 post drill anchor grade 5.6 and base plate thickness of 10mm x 50mm x 200mm. In reference to Figure 3 below, the maximum compression stress experienced at the base plate is equal to: 17.31kN/(0.05mx0.2m) = 1.73MPa. By inspection, the bearing strength of concrete and base plate thickness is satisfied.

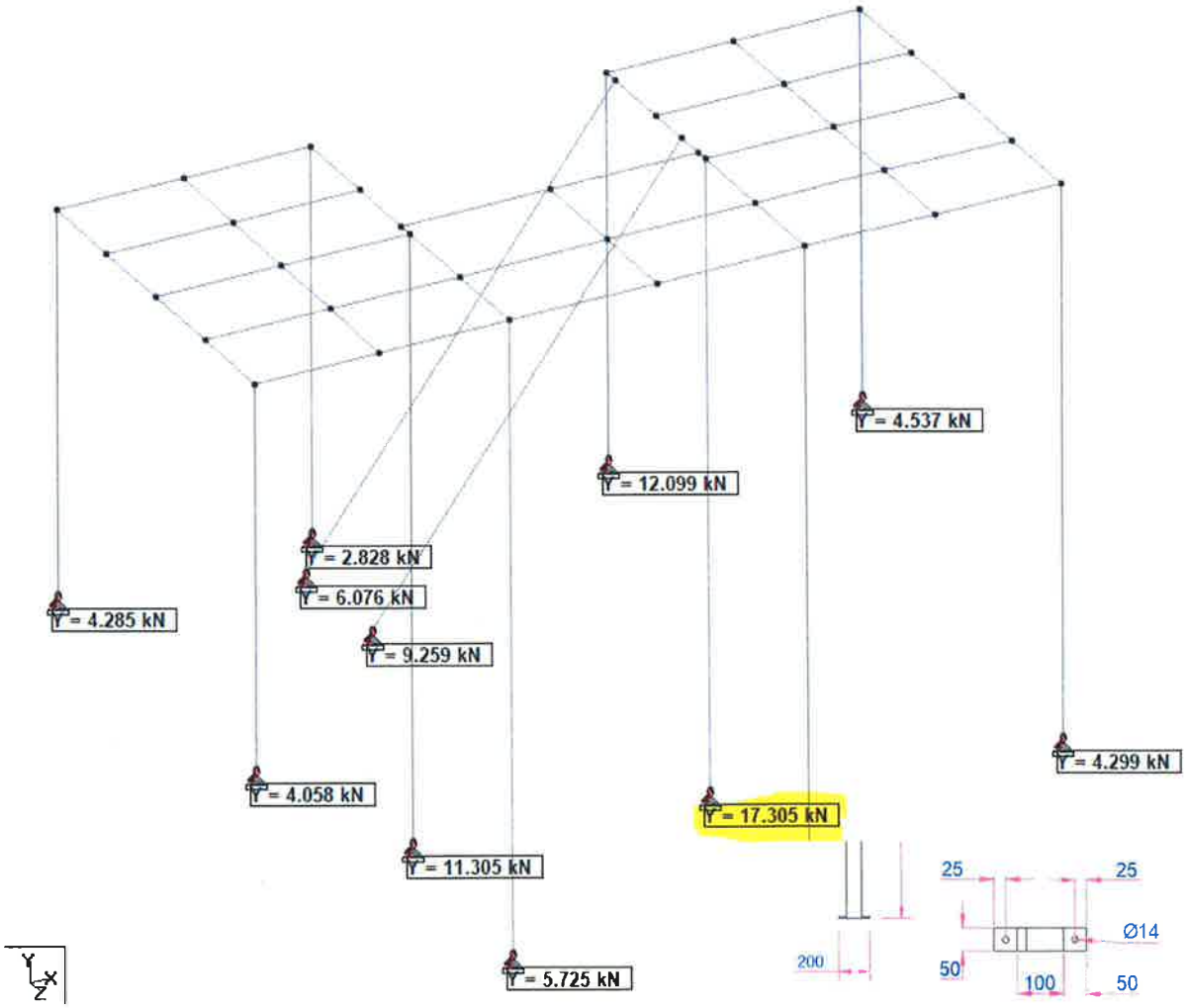


Figure 4. Support Reaction.

5.0 CONNECTION ASSESSMENT

All connections of beam-column and beam-beam connections are fully welded either butt weld of 6mm fillet weld all around. By inspection, no further assessment is required.

6.0 CONCLUSION AND RECOMMENDATION

In reference to Sections 3, 4, and 5 above, the proposed mezzanine framing is adequate to its intended load.

7.0 APPENDIX

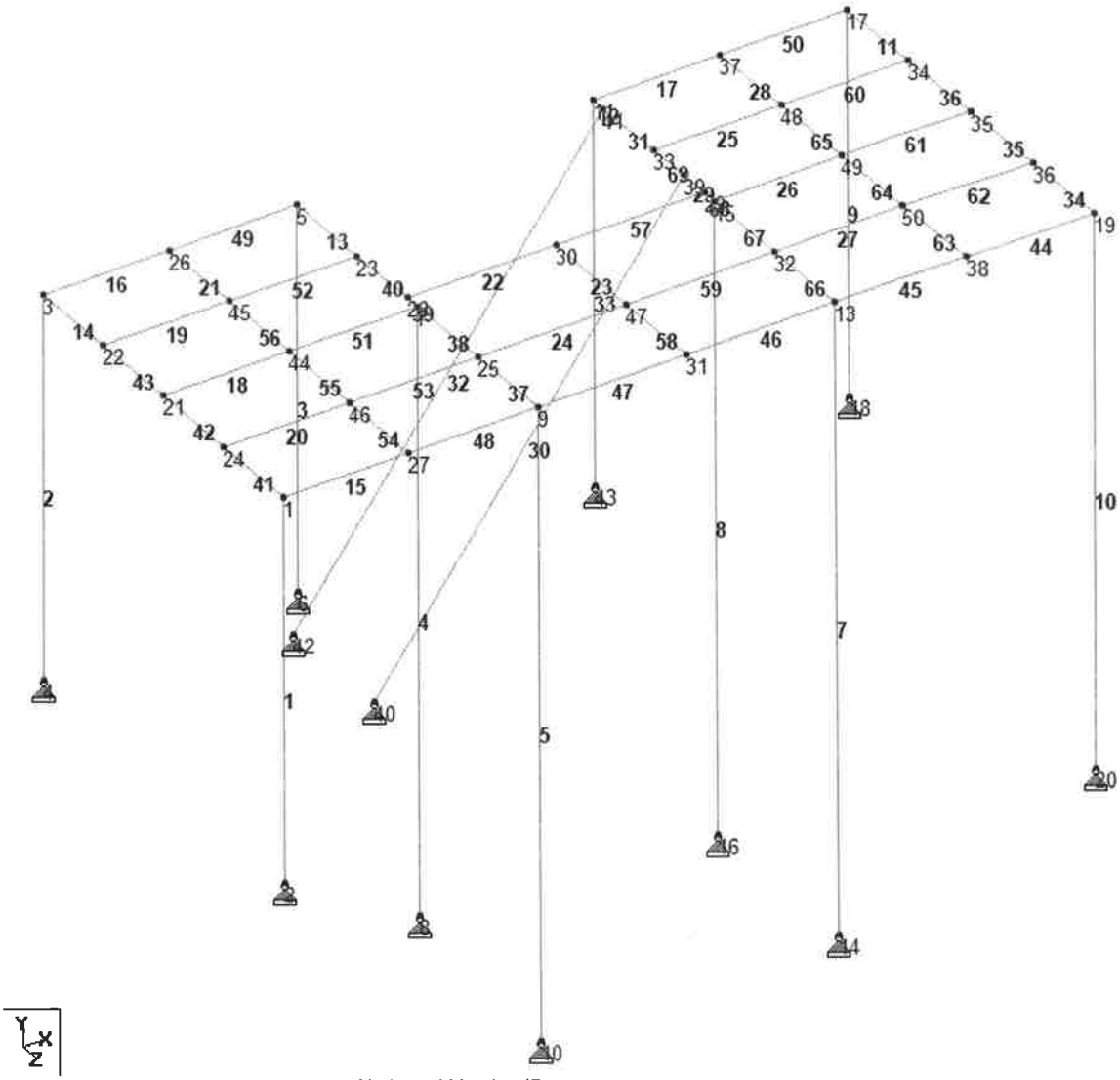
Appendix contains the following:

- Appendix A: STAADPro Analysis
- Appendix B: Detailed Drawings

APPENDIX A

STAADPro ANALYSIS

Puchong Loting Mezzanine Floor Document Storage



Node and Member ID

PAGE NO. 1

```
STAAD SPACE DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF
START JOB INFORMATION
ENGINEER DATE 16-APR-21
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 2.5 2.4; 2 0 0 2.4; 3 0 2.5 0; 4 0 0 0; 5 1.635 2.5 -2.91E-13
6 1.635 0 -0; 7 1.635 2.5 1.2; 8 1.635 -1.45 1.2; 9 1.635 2.5 2.4
10 1.635 -1.6 2.4; 11 3.545 2.5 -2.037E-13; 13 3.545 2.5 2.4
14 3.545 -1.6 2.4; 15 3.545 2.5 1.2; 16 3.545 -1.6 1.2
17 5.18 2.5 5.82E-14; 18 5.18 0 0; 19 5.18 2.5 2.4; 20 5.18 -1.09 2.4
21 0 2.5 1.2; 22 0 2.5 0.6; 23 1.635 2.5 0.6; 24 0 2.5 1.8
25 1.635 2.5 1.8; 26 0.8175 2.5 -1.455E-13; 27 0.805 2.5 2.4
28 1.635 2.5 1.1; 29 3.545 2.5 1.1; 30 2.59 2.5 1.1; 31 2.59 2.5 2.4
32 3.545 2.5 1.8; 33 3.545 2.5 0.6; 34 5.18 2.5 0.6; 35 5.18 2.5 1.2
36 5.18 2.5 1.8; 37 4.3625 2.5 -7.28E-14; 38 4.375 2.5 2.4
39 3.545 2.5 0.9; 40 1.545 -0.19 0.9; 41 3.545 2.5 0.1
42 1.545 -0.19 0.1; 43 3.545 0 -2.619E-13; 44 0.81125 2.5 1.2
45 0.814375 2.5 0.6; 46 0.808125 2.5 1.8; 47 2.59 2.5 1.8
48 4.36563 2.5 0.6; 49 4.36875 2.5 1.2; 50 4.37187 2.5 1.8
MEMBER INCIDENCES
1 1 2; 2 3 4; 3 5 6; 4 7 8; 5 9 10; 7 13 14; 8 15 16; 9 17 18; 10 19 20
11 17 34; 12 11 41; 13 5 23; 14 3 22; 15 1 27; 16 3 26; 17 11 37
18 21 44; 19 22 45; 20 24 46; 21 26 45; 22 28 30; 23 30 47; 24 25 47
25 33 48; 26 15 49; 27 32 50; 28 37 48; 29 39 29; 30 39 40; 31 41 33
32 41 42; 33 11 43; 34 36 19; 35 35 36; 36 34 35; 37 25 9; 38 7 25
39 28 7; 40 23 28; 41 24 1; 42 21 24; 43 22 21; 44 38 19; 45 13 38
46 31 13; 47 9 31; 48 27 9; 49 26 5; 50 37 17; 51 44 7; 52 45 23
53 46 25; 54 46 27; 55 44 46; 56 45 44; 57 30 29; 58 47 31; 59 47 32
60 48 34; 61 49 35; 62 50 36; 63 50 38; 64 49 50; 65 48 49; 66 32 13
67 15 32; 68 29 15; 69 33 39
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05E+08
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2E-05
```

DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF -- PAGE NO. 2

```
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY BRITISH
1 TO 5 7 TO 69 TABLE ST CH100X50X10
CONSTANTS
BETA 180 MEMB 1 7 10 30
BETA 90 MEMB 11 TO 29 31 34 TO 69
MATERIAL STEEL ALL
SUPPORTS
2 4 6 8 10 14 16 18 20 40 42 43 PINNED
LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD
SELFWEIGHT Y -1 LIST ALL
MEMBER LOAD
12 13 22 29 TO 32 40 57 69 UNI GY -0.54
ONEWAY LOAD
YRANGE 2.4 2.6 ONE -0.175 GY
**NOTE** about Floor/OneWay Loads/Weights.
Please note that depending on the shape of the floor you may
have to break up the FLOOR/ONEWAY LOAD into multiple commands.
For details please refer to Technical Reference Manual
Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

MEMBER LOAD
LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD
ONEWAY LOAD
YRANGE 2.4 2.6 ONE -2.5 GY
```

```
  61. MEMBER LOAD
  62. 30 32 UNI GY -2.25
  63. LOAD COMB 3 ULS
  64. 1 1.4 2 1.6
  65. LOAD COMB 4 SLS
  66. 1 1.0 2 1.0
  67. PERFORM ANALYSIS
_

      P R O B L E M   S T A T I S T I C S
      ------------------------------------

   NUMBER OF JOINTS        49  NUMBER OF MEMBERS     68
   NUMBER OF PLATES         0  NUMBER OF SOLIDS       0
   NUMBER OF SURFACES       0  NUMBER OF SUPPORTS     12

      Using 64-bit analysis engine.
_   DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF     -- PAGE NO.   3


      SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=   36/    7/    42 DOF
  TOTAL     PRIMARY LOAD CASES =    2, TOTAL DEGREES OF FREEDOM =    258
  TOTAL LOAD COMBINATION  CASES =    2  SO FAR.
   SIZE OF STIFFNESS MATRIX =      11 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =    12.2/ 312769.8 MB

_


  68. PARAMETER 1
  69. CODE BS5950
  70. LY 1 MEMB 1 TO 5 7 TO 10 33
  71. LZ 1 MEMB 1 TO 5 7 TO 10
  72. PY 275000 ALL
  73. CHECK CODE ALL
_STEEL DESIGN  _


                STAAD.Pro CODE CHECKING - (BSI )
                ********************************

 PROGRAM CODE REVISION V2.13_5950-1_2000
_   DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF     -- PAGE NO.   4

_

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE Noted)

 MEMBER    TABLE       RESULT/  CRITICAL COND/    RATIO/    LOADING/
                FX         MY        MZ     LOCATION
 ==========================================================================

_
    1 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.343       3
              3.71 C      0.81        0.18      -
    2 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.277       3
              3.94 C      0.56        0.41      -
    3 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.090       3
              2.48 C      0.03        0.55      -
    4 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.237       3
             10.75 C      0.00        1.15      -
    5 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.143       3
              5.15 C      0.22        0.22      -
    7 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.098       3
              4.57 C      0.15        0.11      -
    8 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.293       3
```

```
            16.73 C      0.12      1.02      -
 9 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.342      3
            4.19 C      0.69      0.54      -
10 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.293      3
            3.80 C      0.56      0.44      -
11 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.266      3
            0.07 C      0.69      0.10      -
12 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.258      3
            0.33 C      0.65      0.16      -
13 ST CH100X50X10  PASS   BS-4.9          0.111      3
            0.00 T      0.30      0.01     0.00
14 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.288      3
            0.23 C      0.77      0.00      -
15 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.140      3
            0.05 C      0.37      0.01      -
16 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.176      3
            0.17 C      0.47      0.00      -
17 ST CH100X50X10  PASS   BS-4.9          0.185      3
            0.00 T      0.48      0.09     0.00
18 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.193      3
            0.04 C      0.52      0.01      -
19 ST CH100X50X10  PASS   BS-4.9          0.137      3
            0.00 T      0.37      0.00     0.00
20 ST CH100X50X10  PASS   BS-4.9          0.123      3
            0.00 T      0.33      0.02     0.00
21 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.242      3
            0.01 C      0.65      0.01      -
22 ST CH100X50X10  PASS   BS-4.9          0.316      3
            0.00 T      0.84      0.02     0.00
```


ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

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MEMBER    TABLE       RESULT/  CRITICAL COND/   RATIO/    LOADING/
                FX        MY        MZ     LOCATION
=========================================================================

23 ST CH100X50X10  PASS   BS-4.9          0.032      1
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24 ST CH100X50X10  PASS   BS-4.9          0.348      3
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25 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.108      3
            0.69 C      0.25      0.13      -
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            0.49 C      1.24      0.05      -
27 ST CH100X50X10  PASS   BS-4.9          0.399      3
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28 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.286      3
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29 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.288      3
            0.57 C      0.75      0.09      -
30 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.801      3
            0.16 C      0.25      3.71      -
31 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.333      3
            0.38 C      0.87      0.08      -
32 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.677      3
            0.25 C      0.16      3.64      -
33 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.217      3
           11.75 C      0.35      0.28      -
34 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.290      3
            0.06 C      0.77      0.04      -
35 ST CH100X50X10  PASS   BS-4.9          0.466      3
            0.00 T      1.24      0.06     0.00
36 ST CH100X50X10  PASS   BS-4.9          0.466      3
            0.00 T      1.24      0.04     0.00
37 ST CH100X50X10  PASS   BS-4.8.3.3.1     0.389      3
```

```
            0.01 C      1.04      0.02      -
38 ST CH100X50X10  PASS   BS-4.9          0.396      3
            0.00 T      1.06      0.03      0.00
39 ST CH100X50X10  PASS   BS-4.9          0.335      3
            0.00 T      0.90      0.02      0.00
40 ST CH100X50X10  PASS   BS-4.9          0.180      3
            0.00 T      0.48      0.02      0.00
41 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.306      3
            0.30 C      0.82      0.01      -
42 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.450      3
            0.26 C      1.20      0.01      -
43 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.449      3
            0.24 C      1.21      0.00      -
44 ST CH100X50X10  PASS   BS-4.9          0.168      3
            0.00 T      0.44      0.04      0.00
  DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF     -- PAGE NO.   6


ALL UNITS ARE - KN   METE (UNLESS OTHERWISE Noted)

MEMBER    TABLE       RESULT/  CRITICAL COND/    RATIO/    LOADING/
            FX          MY        MZ      LOCATION
=============================================================================


45 ST CH100X50X10  PASS   BS-4.9          0.149      3
            0.00 T      0.39      0.05      0.00
46 ST CH100X50X10  PASS   BS-4.9          0.188      3
            0.00 T      0.50      0.03      0.00
47 ST CH100X50X10  PASS   BS-4.9          0.143      3
            0.00 T      0.38      0.02      0.00
48 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.223      3
            0.01 C      0.60      0.00      -
49 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.198      3
            0.20 C      0.53      0.01      -
50 ST CH100X50X10  PASS   BS-4.9          0.210      3
            0.00 T      0.54      0.09      0.00
51 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.505      3
            0.09 C      1.36      0.01      -
52 ST CH100X50X10  PASS   BS-4.9          0.130      3
            0.00 T      0.35      0.01      0.00
53 ST CH100X50X10  PASS   BS-4.9          0.344      3
            0.00 T      0.92      0.01      0.00
54 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.148      3
            0.02 C      0.40      0.01      -
55 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.149      3
            0.03 C      0.40      0.02      -
56 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.242      3
            0.03 C      0.65      0.00      -
57 ST CH100X50X10  PASS   BS-4.9          0.315      3
            0.00 T      0.84      0.03      0.00
58 ST CH100X50X10  PASS   BS-4.9          0.031      1
            0.00 T      0.08      0.01      0.00
59 ST CH100X50X10  PASS   BS-4.9          0.399      3
            0.00 T      1.07      0.04      0.00
60 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.122      3
            0.28 C      0.31      0.08      -
61 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.194      3
            0.19 C      0.52      0.01      -
62 ST CH100X50X10  PASS   BS-4.9          0.131      3
            0.00 T      0.34      0.06      0.00
63 ST CH100X50X10  PASS   BS-4.9          0.140      3
            0.00 T      0.36      0.07      0.00
64 ST CH100X50X10  PASS   BS-4.9          0.168      3
            0.00 T      0.44      0.08      0.00
65 ST CH100X50X10  PASS   BS-4.9          0.277      3
            0.00 T      0.74      0.04      0.00
66 ST CH100X50X10  PASS   BS-4.8.3.3.1    0.328      3
```

```
              0.19 C       0.88        0.02      -
  DXF IMPORT OF HYPERWAVE - HSP PUCHONG LOTING 3D.DXF     -- PAGE NO.   7

ALL UNITS ARE - KN   METE (UNLESS OTHERWISE Noted)

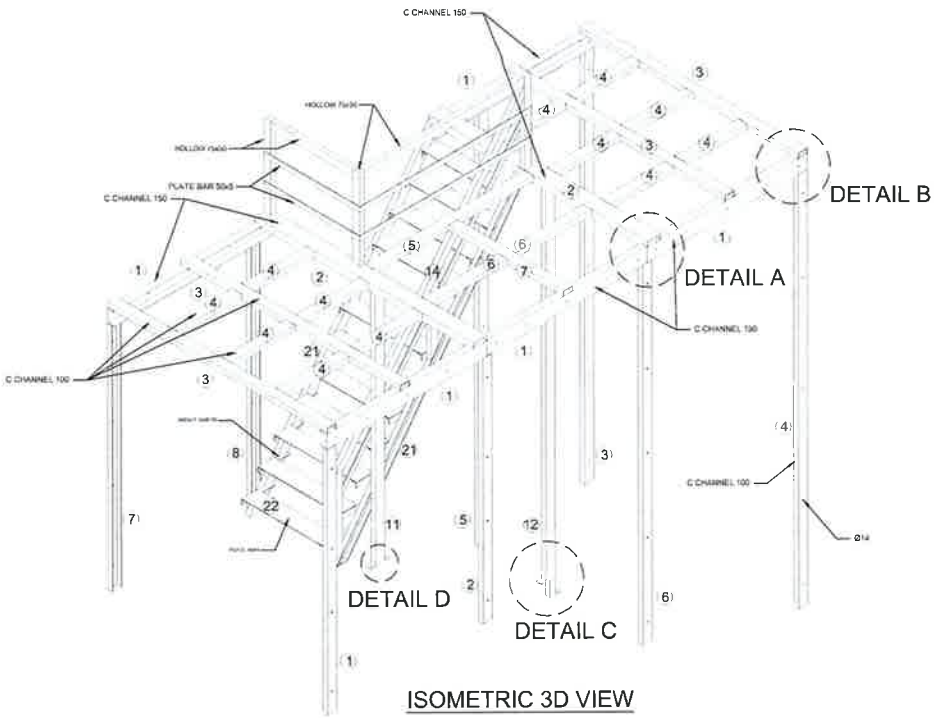
MEMBER    TABLE       RESULT/  CRITICAL COND/    RATIO/    LOADING/
                FX        MY        MZ     LOCATION
=========================================================================

  67 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.562      3
              0.40 C       1.51        0.02      -
  68 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.610      3
              0.52 C       1.63        0.04      -
  69 ST CH100X50X10  PASS    BS-4.8.3.3.1     0.352      3
              0.64 C       0.88        0.24      -

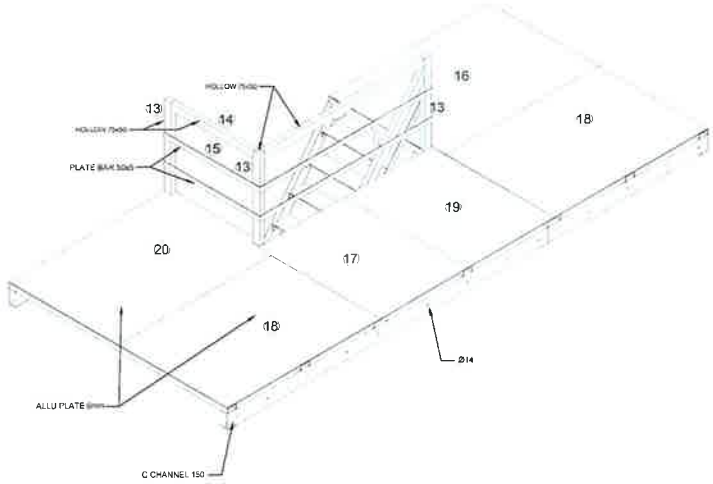
 ************** END OF TABULATED RESULT OF DESIGN **************
```

APPENDIX B

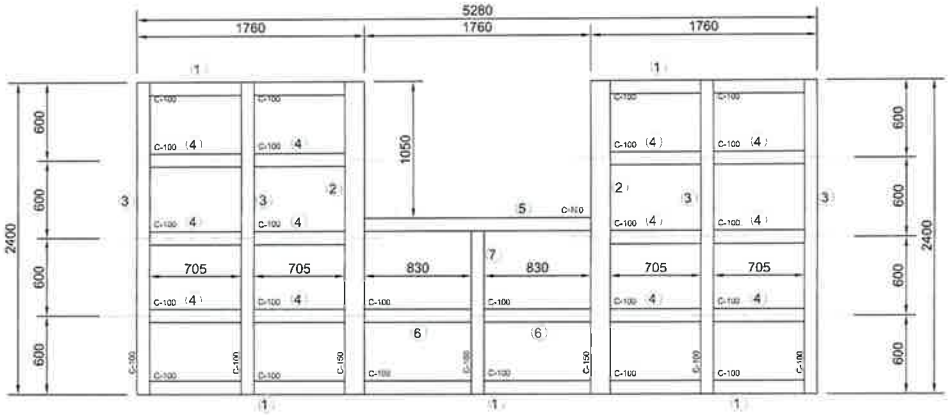
DETAILED DRAWING



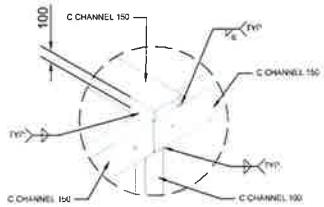
ISOMETRIC 3D VIEW



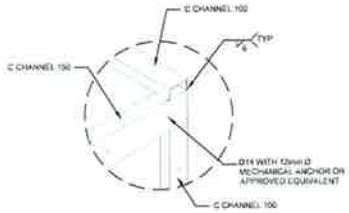
ISOMETRIC 3D VIEW
FLOOR PLAN



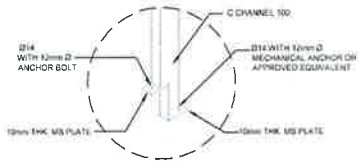
FLOOR FRAMING PLAN



DETAIL A



DETAIL B



DETAIL C

DETAIL D

TYPICAL CONNECTION DETAIL

NOTES

REFERENCE DRAWING

CLIENT

Hyperwave Systems Engineering Sdn Bhd

SIGNATURE

CONSULTANT

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BEM REGISTERED NUMBER SP 1545

231, 2ND FLOOR, PHASE III,
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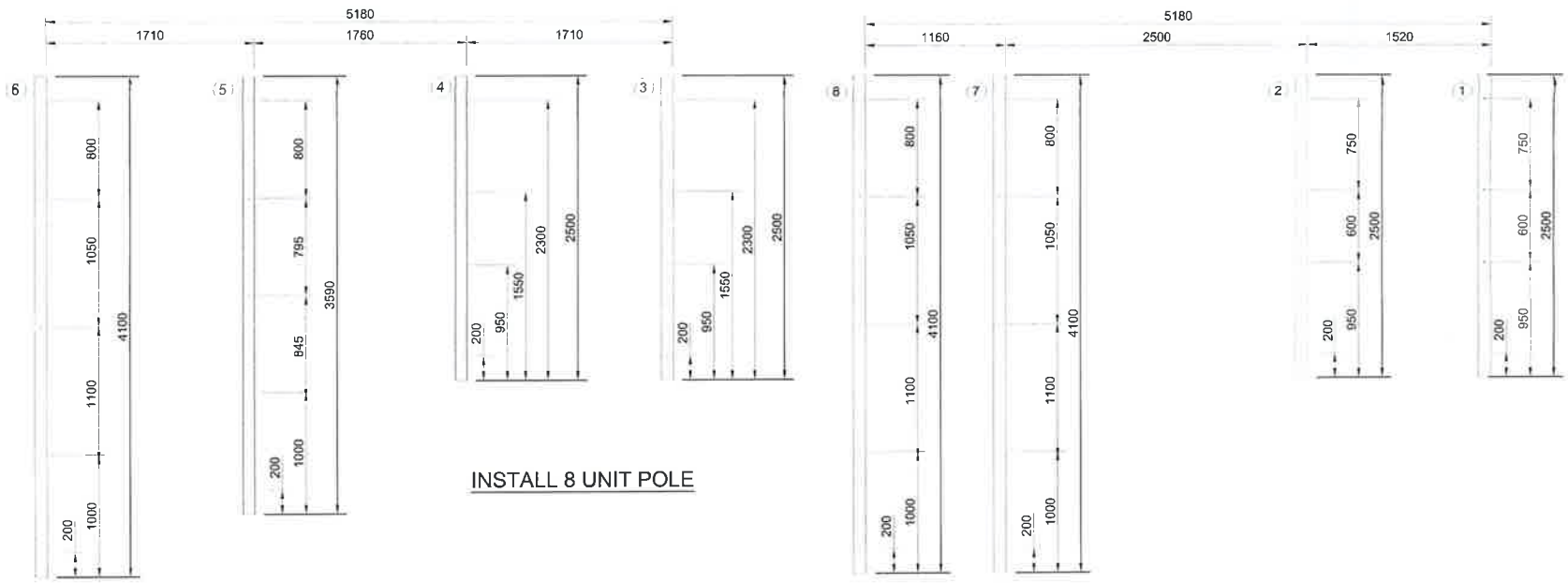
PROJECT TITLE

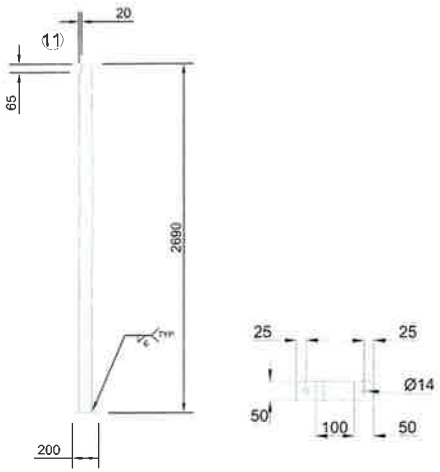
HYPERWAVE PUCHONG
PLATFORM DETAIL

SCALE AS SHOWN

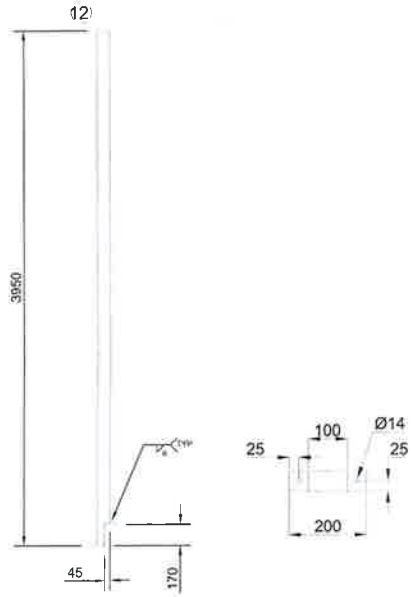
DESIGNED BY	CHECKED BY
DRAWN BY	APPROVED BY
JOB NO	DATE APL 2021
DRAWING NO HSP-HYW-001	SHEET NO 1/5 REVISION 0

PROFESSIONAL ENGINEER WITH PRACTISING CERTIFICATE
CIVIL
Ir. HAW SENG POH
C116525
MALAYSIA





NAME : POLE 4
MATERIAL : HOLLOW 100x50
QTY : 1



NAME : POLE 5
MATERIAL : HOLLOW 100x50
QTY : 1

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NOTES:

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PROJECT TITLE

DRAWING TITLE
HYPERWAVE PUCHONG
PLATFORM DETAIL

SCALE	AS SHOWN
DESIGNED BY	CHECKED BY
DRAWN BY	APPROVED BY
JOB NO	DATE APL 2021
DRAWING NO HSP-HYW-001	SHEET NO 2/5 REVISION 0

NAME : BASE WALL
MATERIAL : C CHANNEL 150
QTY : 5

NAME : BASE Y 100
MATERIAL : C CHANNEL 100
QTY : 12

NAME : BASE X 100
MATERIAL : C CHANNEL 100
QTY : 2

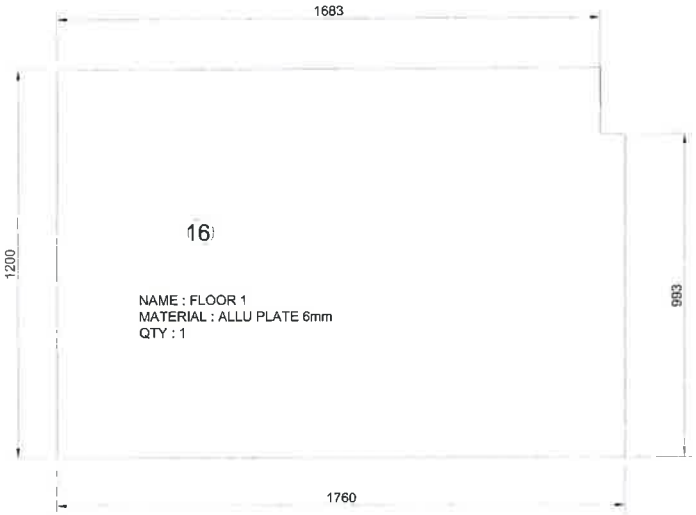
NAME : BASE Y 150
MATERIAL : C CHANNEL 150
QTY : 2

NAME : BASE X 100
MATERIAL : C CHANNEL 100
QTY : 1

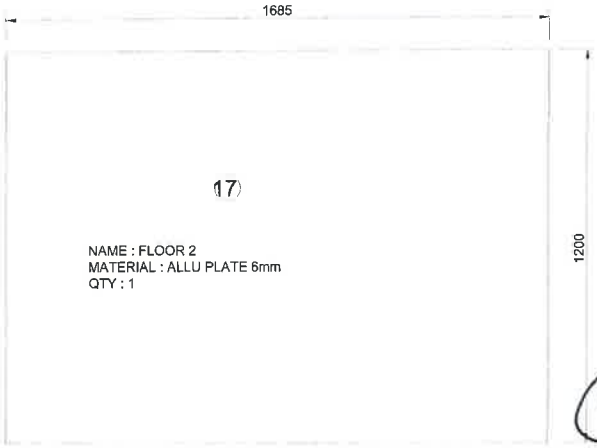
NAME : BASE X 100
MATERIAL : C CHANNEL 100
QTY : 4

NAME : BASE X 100
MATERIAL : C CHANNEL 100
QTY : 1

NAME : FLOOR 3
MATERIAL : ALLU PLATE
6mm
QTY : 2



NAME : FLOOR 1
MATERIAL : ALLU PLATE 6mm
QTY : 1



NAME : FLOOR 2
MATERIAL : ALLU PLATE 6mm
QTY : 1

NOTES

REMARKS

REFERENCE DRAWING

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SIGNATURE

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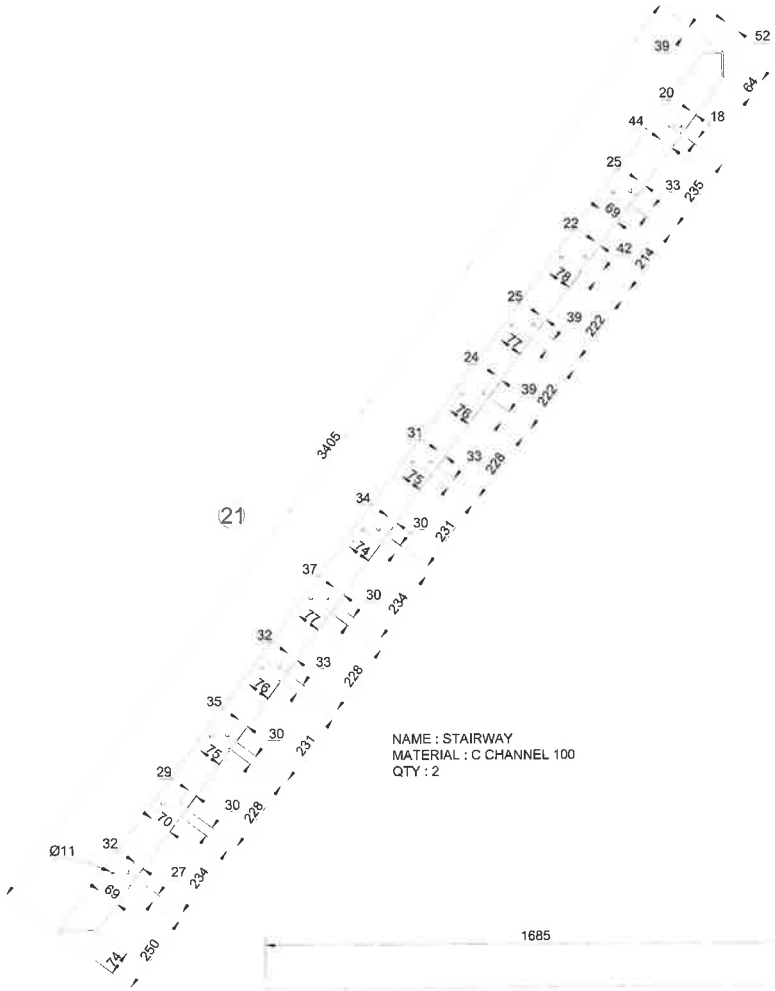
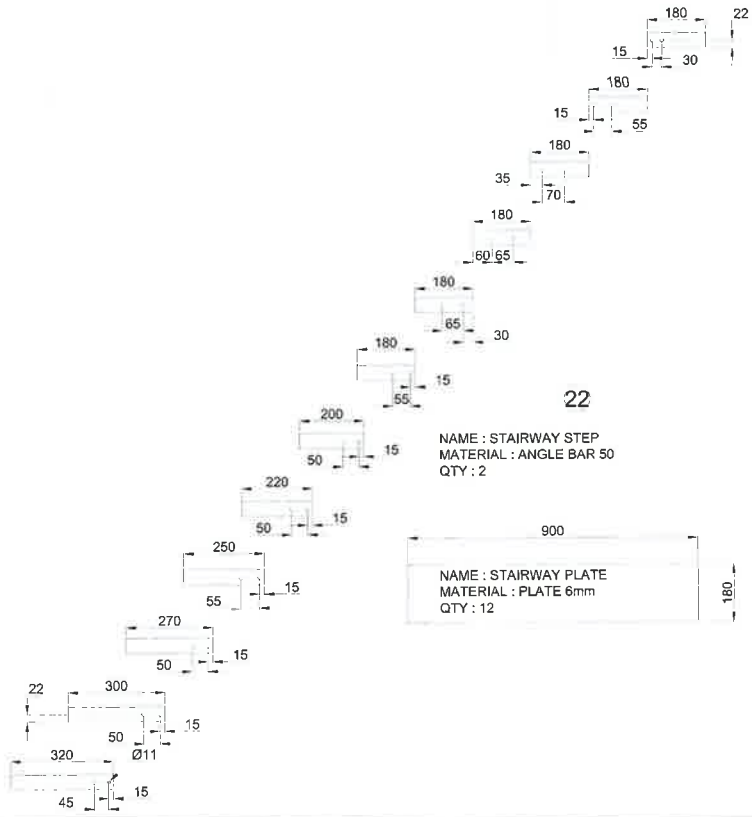
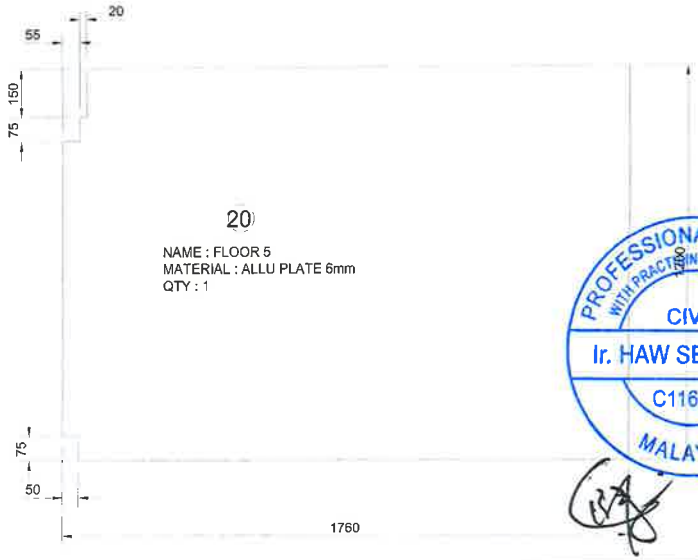
HYPERWAVE PUCHONG
PLATFORM DETAIL

SCALE : AS SHOWN		
DESIGNED BY	CHECKED BY	
DRAWN BY	APPROVED BY	
JOB NO	DATE APL 2021	
DRAWING NO : HSP-HYW-001	SHEET NO 3/5	REVISION 0







NOTES

REFERENCE DRAWING

CLIENT
Hyperwave Systems Engineering Sdn Bhd

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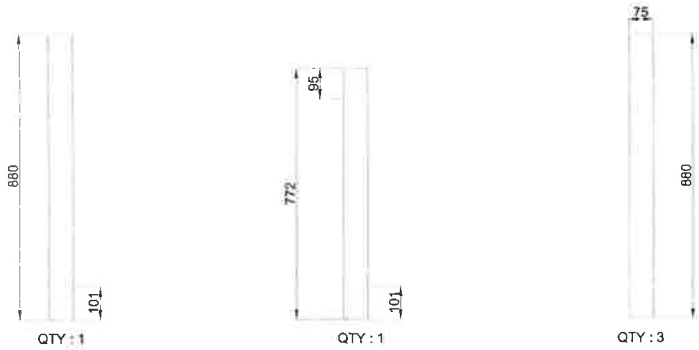
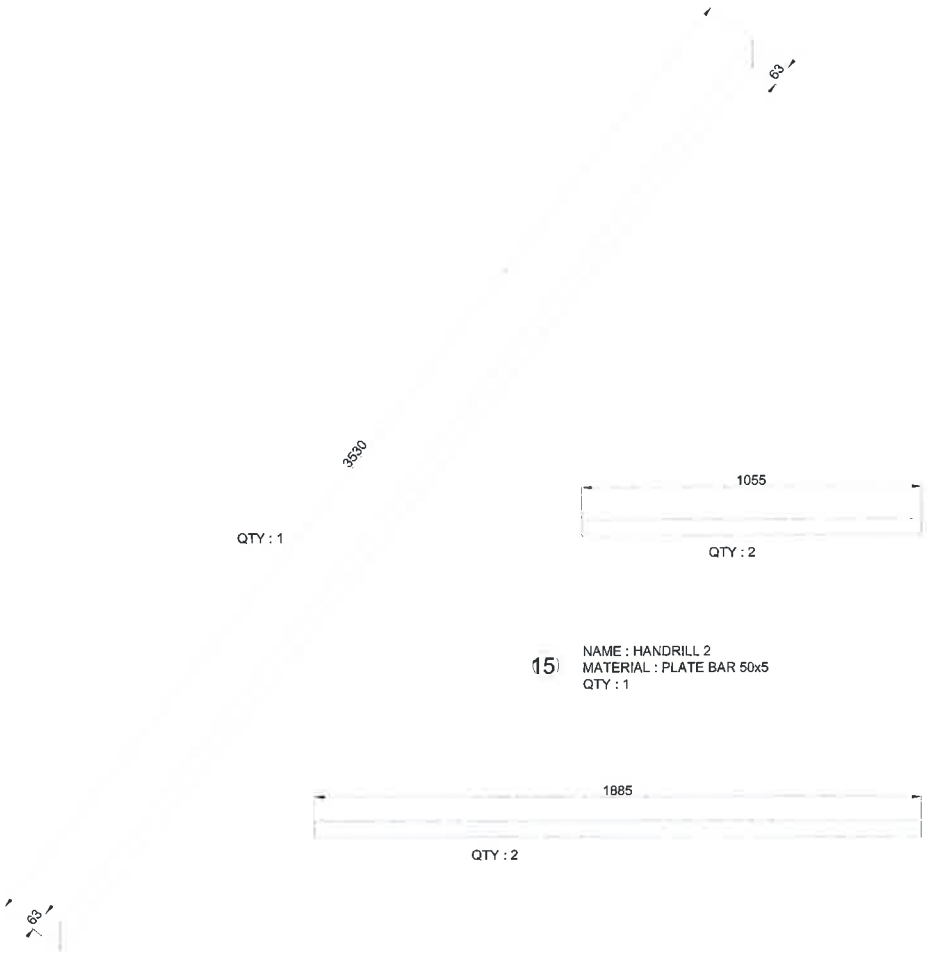
DRAWING TITLE
HYPERWAVE PUCHONG PLATFORM DETAIL

SCALE: AS SHOWN		
DESIGNED BY	CHECKED BY	
DRAWN BY	APPROVED BY	
JOB NO	DATE APL 2021	
DRAWING NO HSP-HYW-001	SHEET NO 4/5	REVISION 0



13) NAME : POLE 6
MATERIAL : HOLLOW 75x50



15) NAME : HANDRILL 2
MATERIAL : PLATE BAR 50x5
QTY : 1

14) NAME : HANDRILL 1
MATERIAL : HOLLOW 75x50
QTY : 1

NOTES

REFERENCE DRAWING

CLIENT
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TITLE

DRAWING TITLE
HYPERWAVE PUCHONG PLATFORM DETAIL

SCALE AS SHOWN

DESIGNED BY	CHECKED BY
DRAWN BY	APPROVED BY
JOB NO	DATE APL 2021
DRAWING NO HSP-HYW-001	SHEET NO. 5/5
	REVISION 0